International Stablecoin Flows, Capital Controls and Currency Crises

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International Monetary Fund

CBER Crafting the Cryptoeconomy Conference 2026

The views expressed herein are those of the author and do not necessarily represent the views of the IMF, the IMF Executive Board, or IMF management.

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- Official statistics do not yet comprehensively measure stablecoin flows and holdings.

This paper

What we do

Estimate stablecoin flows and holdings for 188 countries

What we find

▸ Related literature

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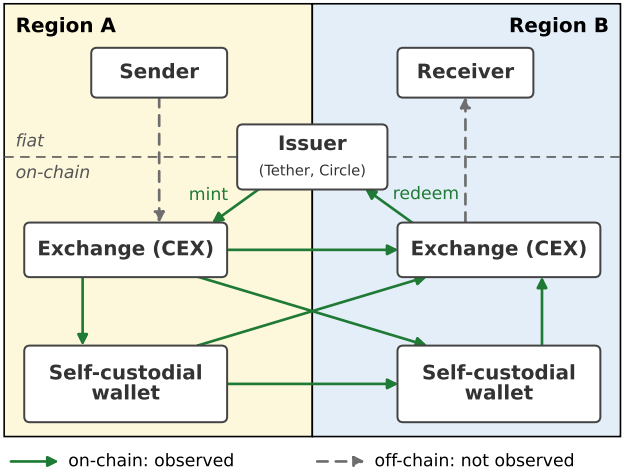
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What we do	What we find
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Event studies around currency crises	Currency crises: stablecoin holdings up **roughly 75 percent**

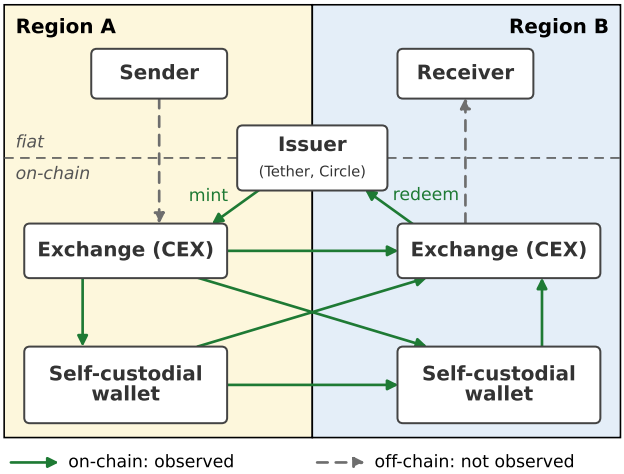
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How an international stablecoin flow works



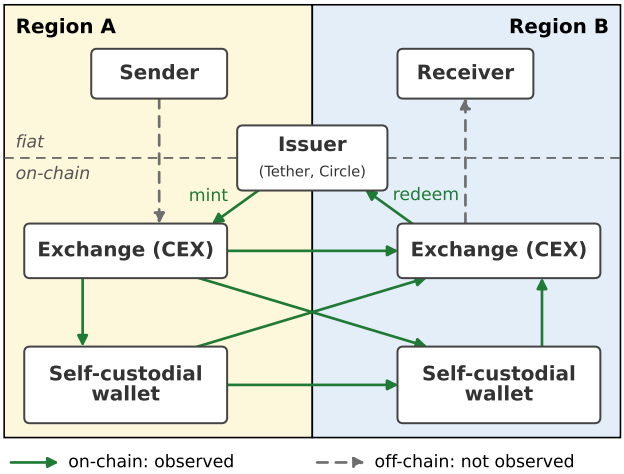
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How an international stablecoin flow works



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- The blockchain records addresses, not where they are.

From blockchain data to a country-level panel

1. Label wallets whose region is known ▸ Labels
   - Using domain names
   - Using regional exchanges

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4. Split regions into countries ▸ Example
   - Using web traffic to the exchanges a wallet uses
   - China, Hong Kong SAR and Macao SAR: a separate machine learning model

USDT and USDC on seven blockchains, 2018–2025 ▸ Coverage

Validation

Daylight saving time	North America shifts with US clocks, Asia does not	▸ Figure

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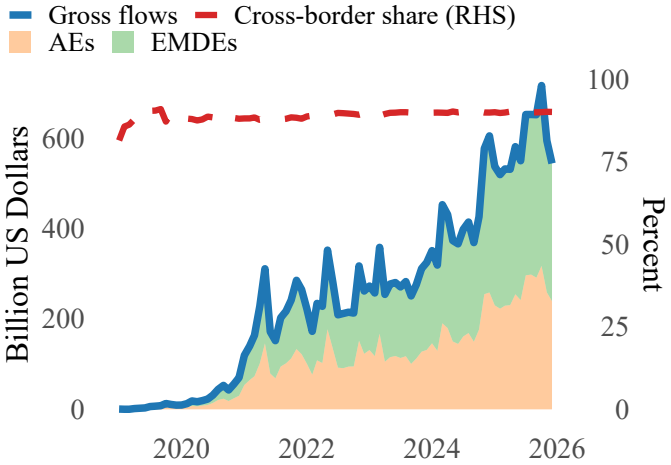
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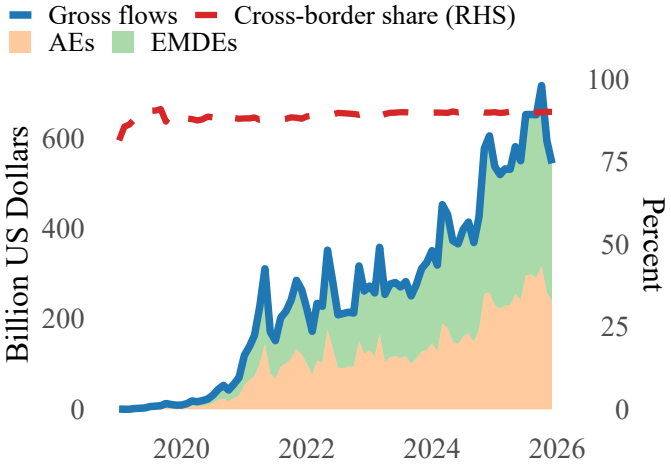
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Binance, China	About 20 percent disclosed (2023), 19 ours	▸ More

Fact 1: large, fast-growing, and cross-border



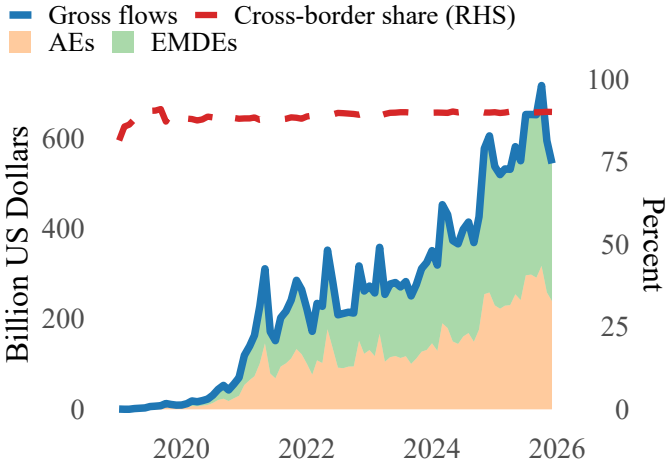
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- About 90 percent crosses borders.
- Median transfer of $300

Fact 2: unlike traditional payments, stablecoin flows center on EMDEs

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  - In SWIFT, 15 percent involve EMDEs, 85 percent between AEs. (Basu et al. 2026)

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  - Venezuela 89, Ukraine 58, Bolivia 38 percent of GDP in 2025 ▸ Top 20

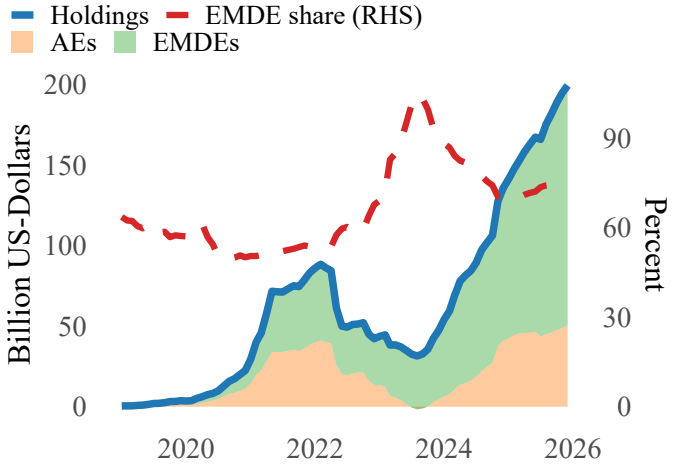
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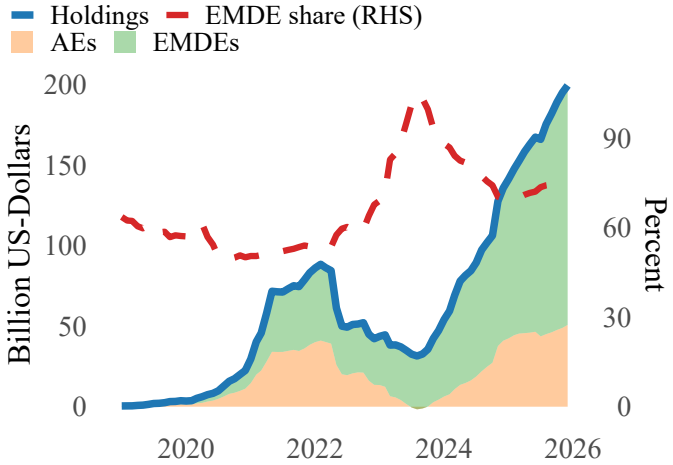
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- Flows are larger in EMDEs with more volatile exchange rates.
  - One standard deviation more volatility, 9.7 percent larger inflows

Fact 3: three quarters of stablecoins are held in EMDEs



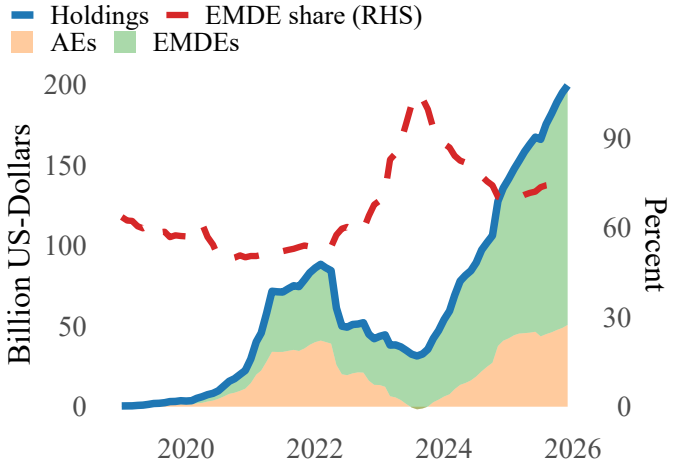
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- About $200 billion at the end of 2025
- All 20 largest holders relative to GDP are EMDEs.
- Venezuela 5.6, Ukraine 4.1, Pakistan 1.4, Nigeria 1.4 percent of GDP

▸ Top 20 ▸ In dollars

Capital flow management measures: data and design

- **Data.** Monthly count of CFMs by country and direction (Bergant et al. 2026)

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- **Controls.** Countries with no activation of their own within ± 6 months, about 140 per event

▸ Other drivers

Two stacked event-study regressions

Baseline: one path for all 217 activations

$$\ln\left(SC_{c,t}\right) = \sum_{\substack{k=-6 \\ k\neq -1}}^{6} \beta_k \, 1\{t - t_e = k\} \times Treat_{c,e} + \alpha_{e,k} + \delta_{e,c} + \varepsilon_{c,t,e}$$

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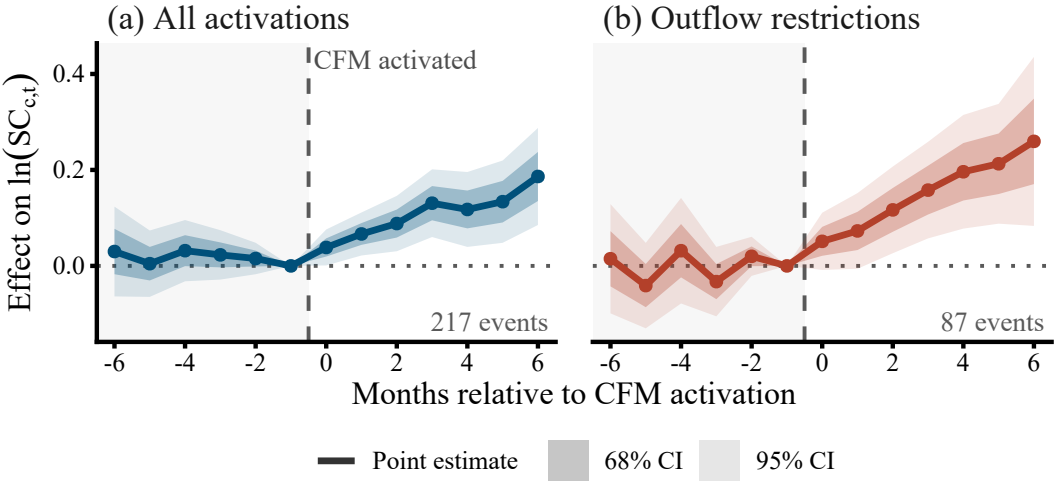
By direction: one path for each direction of the tightened restriction

$$\ln(SC_{c,t}) = \sum_{d\in\mathcal{D}} \sum_{\substack{k=-6 \\ k\neq -1}}^{6} \beta_k^d 1\{t - t_e = k\} \times Treat_{c,e} \times 1\{d(e) = d\} + \alpha_{e,k} + \delta_{e,c} + \varepsilon_{c,t,e}$$

$\mathcal{D} = \{\text{outflow, inflow, neutral, several}\}$, in one joint regression with the same fixed effects and controls

$SC_{c,t}$ is cross-border stablecoin inflows. Inflows and outflows move together, so we read the results as effects on activity.

Stablecoin activity rises after CFM activations



▸ Estimates ▸ 12 months

Unpacking the outflow response

- **Size.** About **$205 million a month** of additional cross-border activity for the median outflow tightening

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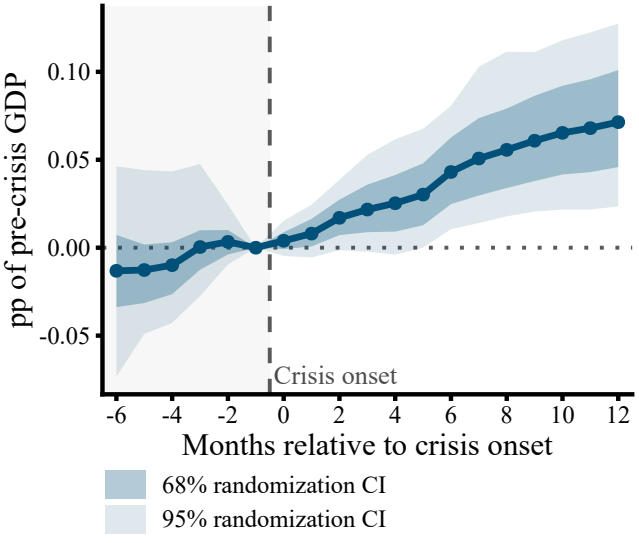
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- **Robustness.** Not macroeconomic stress: no pre-trend, unchanged with inflation and FX volatility controls, larger without crisis countries ▸ Details

Stablecoin holdings rise after currency crises

- 21 currency crises, 2021–2025 (Laeven–Valencia criterion)
  ▸ Events
- Same stacked event-study design as for CFMs
- Holdings in percent of pre-crisis GDP
- Controls: EMDEs never in crisis
  - 15 per crisis, same region
  - Similar holdings, holdings volatility, and GDP beforehand



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- **Robustness.** Dating, estimators, CFM controls, without Russia and Ukraine ▸ Table

Conclusion

- A methodology to estimate stablecoin flows and holdings and a dataset that allocates $21.4 trillion of stablecoin transfers across 188 countries
- Our estimates show stablecoins operating mainly as a dollar rail for EMDEs.
- We show that stablecoin activity rises after outflow restrictions, and that holdings rise after currency crises. Both responses persist.

Paper

Data

Backup

Related literature

Back

Strand	What we add
Wallet geography from on-chain data Athey et al. (2016), Cong et al. (2026)	437,775 labeled wallets, and bilateral flows at the country level
International crypto flows Cerutti et al. (2025), Auer et al. (2025), Hofmann et al. (2026)	A country panel that attributes wallets instead of exchange websites, and measures holdings
Capital controls and crypto von Luckner et al. (2023, 2024), Hu et al. (2021)	First event-study evidence for stablecoins, around 217 activations
Crisis-driven dollarization Reinhart et al. (2014), Levy Yeyati (2021), Azar et al. (2026)	A monthly dollarization stock, so the path around crisis onset can be estimated

Covered transfers

Back

We start from all USDT and USDC transfers on the seven chains and drop five classes.

1. Either side is a smart contract. This removes decentralized-finance activity.
2. Both sides are the same centralized exchange. These are operational movements between its own wallets.
3. A self-custodial wallet with fewer than 50 initiated transactions, for which the model is not trained.
4. An automated trading program, institutional trading firm, liquidity provider, or OTC desk.
5. A self-custodial wallet that has never interacted with a centralized exchange, for which the country split is undefined.

- What remains is simple transfers involving exchanges and sufficiently active self-custodial wallets.
- Implied holdings cover 78 percent of stablecoins outstanding at the end of 2025. Of the remainder, 8 percentage points sit on chains outside the sample and 14 with entities we do not cover.

Step 1: training labels from two sources

Back

- **Domain names.** A language model reads almost 6 million wallet names for linguistic and cultural markers.
  - `pijiu.eth`: *pijiu* is Chinese for "beer"
- **Regional exchanges.** Wallets with more than 90 percent of their exchange transactions at exchanges serving one region.

Region	Domain	Exchange	Total
Africa and Middle East	19,731	21,123	40,854
Asia and Pacific	129,516	90,472	219,988
Europe	94,937	40,466	135,403
Latin America and Caribbean	3,905	5,621	9,526
North America	23,569	8,435	32,004
Total	271,658	166,117	**437,775**

Step 2: classification model and identifying assumption

Back

- Gradient boosted trees, trained on the on-chain behavior of labeled wallets
  - Time-of-day activity
  - (Non)adherence to daylight saving time
  - Exchanges, tokens, and smart contracts used
- Output: a probability distribution over five regions for every self-custodial wallet with at least 50 transactions

Identifying assumption. Conditional on the features, labeled and unlabeled wallets of a region follow the same data generating process.

It fails if wallets that buy domain names or use regional exchanges behave differently from other wallets in their region. It cannot be tested for wallets that are never labeled.

Step 4: from regions to countries

Back

- Each region is split across countries with the web traffic of the exchanges a wallet uses.
- **China** has a dedicated model, as exchange websites are blocked and VPN use distorts traffic.
- The traffic assumptions apply within a region only. VPN use cannot move volume across regions.
- Each transfer is then allocated to country pairs.

Formulas

A worked example

$P(\text{Asia and Pacific}) = 0.8$

Indonesia's share of web traffic: 0.75

$P(\text{Indonesia}) = 0.8 \times 0.75 = 0.6$

The wallet sends \$100 to a wallet with $P(\text{United States}) = 0.5$.

Indonesia to United States: \$30

Allocation formulas

Back

Regional share of exchange x at hour t on blockchain b, from its stablecoin traffic V with located self-custodial wallets:

$$P_x(r \mid t, b) = \frac{V_{x,r,t,b}}{\sum_{r'} V_{x,r',t,b}}$$

Country split with web traffic $W_{c,x,t}$ from country c, within region r only:

$$P_x(c \mid t, b) = \frac{W_{c,x,t}}{\sum_{c' \in r} W_{c',x,t}} \cdot P_x(r \mid t, b)$$

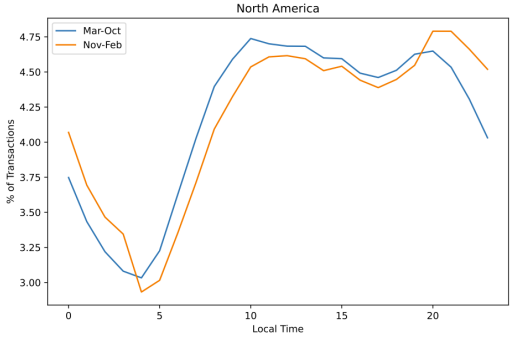
Wallet country probabilities, a volume-weighted average over the exchanges the wallet has used:

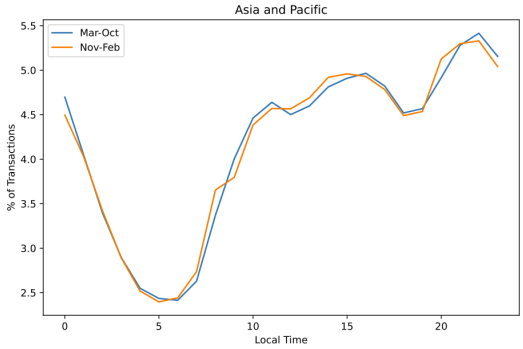
$$P_w(c) = \frac{\sum_x \sum_{t,b} V_{w,x,t,b} \cdot P_x(c \mid t, b)}{\sum_x \sum_{t,b} V_{w,x,t,b}}$$

Transfer of value v from w to w': the amount assigned to the pair (c, c') is $v \cdot P_w(c) \cdot P_{w'}(c')$. Exchange endpoints use $P_x(\cdot \mid t, b)$. Summing over pairs returns v.

Validation of the labels: daylight saving time

Back

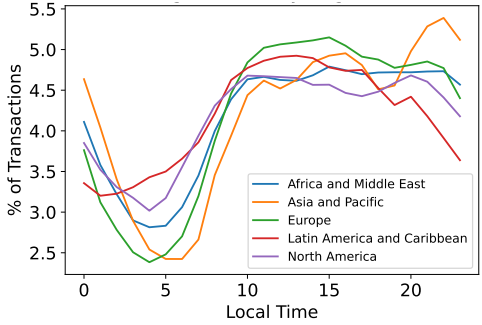




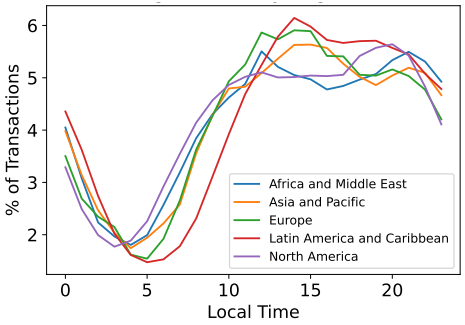
- Hourly activity of labeled wallets, with and without US daylight saving time Activity profiles
- Wallets labeled North America shift by about one hour. Wallets labeled Asia and Pacific do not.

Activity profiles of labeled wallets in region-local time

Back



Labeled through domain names



Labeled through regional exchanges

Share of each wallet group's transactions by hour of day, with UTC converted to a representative local time for the assigned region.

Validation of the model: held-out wallets

Back

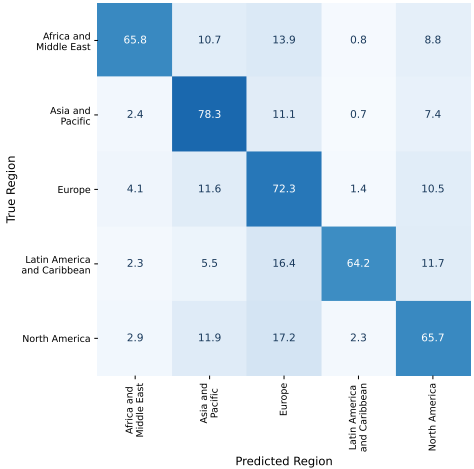
Test	Wallets	Accuracy	Benchmark
Five regions, held-out wallets	43,768	**74.1**	20.0
of which labeled by domain name		58.3	47.5
Train on domain names, test on exchange-labeled wallets	166,117	**91.5**	54.5
China against the rest of Asia and Pacific	21,997	83.1	50.0

Accuracy and benchmark in percent.

- In the flow allocation, 62 cents of every dollar sent by the average held-out wallet go to its true region. Confusion matrix

Confusion matrix on held-out wallets

Back

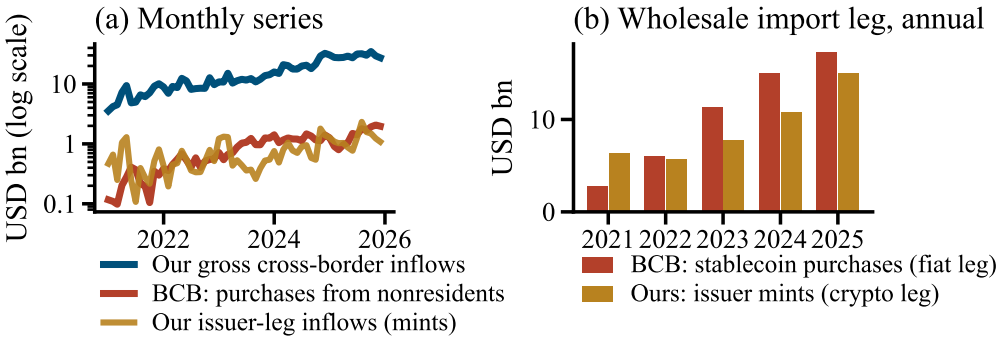


Model	Acc.	Major.	Dom.	CEX
Stage 1 (three groups)	77.1	50.3	63.1	99.9
Stage 2: Africa–ME vs. Europe	86.7	76.8	79.6	99.9
Stage 2: Lat. Am. vs. North Am.	89.1	77.0	83.8	100.0
China vs. rest of Asia–Pacific	81.9	80.0	69.1	100.0
Combined five regions	74.1	50.3	58.3	99.9
Cross-source test	91.5	54.5		

Percent of held-out wallets. Major. is the accuracy of always guessing the largest class. Dom. and CEX split accuracy by labeling source.

External check: Brazil's balance of payments statistics

Back



- Brazil's central bank records stablecoin purchases from FX contracts, with no inputs shared with our methodology.
- 2021–2025: $52.3 billion (central bank) against $45.5 billion of mints in our estimates. **Within 15 percent.** Net flows

Brazil: reconciling net flows

Back

USD billion, 2021–2025	BCB	Ours
Wholesale imports	52.3	45.5
Wholesale exports	1.9	34.9
Net flows with other countries	–	2.0
Net	50.4	12.6

BCB: purchases and sales of stablecoins by residents, from foreign exchange contracts. Ours: issuer mints into Brazil, redemptions from Brazil, and bilateral net inflows summed over partners.

- The gap in net flows is the redemption leg.
- Little of it should appear in the foreign exchange records.
  - A global platform that redeems is not a Brazilian resident, so no contract arises.
  - A Brazilian exchange receives dollars that need not be converted into reais.
- BCB net purchases less our redemptions leave $15.5 billion, within $3 billion of our accumulation estimate.

External checks: exchange disclosures and Chainalysis

Back

Internal figures from exchanges

Share (%)	Disclosed	Ours
Binance, US, 2020	16–19.9	16.8–24.8
Binance, China, 2023	≈20	19.3
KuCoin, US, 2019–23	20–50	18.2

Disclosed in CFTC complaints and in reporting by the Wall Street Journal.

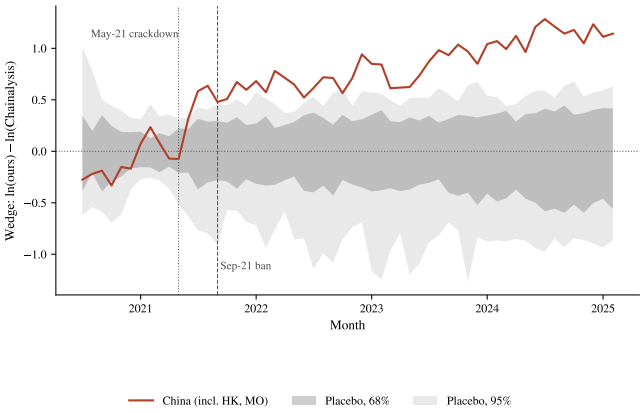
Chainalysis

- Same cross-section: correlation of 0.93 in gross flows
- Different time series: 0.26 after country and month fixed effects
- Our levels are 2.6 times larger, and 14 times for China China
- No issuer legs, so no holdings

The event studies use time-series variation and holdings. On the Chainalysis series, neither design detects a significant response.

Chainalysis comparison: the China wedge

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- Log ratio of our estimate of China's gross flows to the Chainalysis estimate.
- Unremarkable before May 2021.
- It jumps by about 0.6 log points between the May 2021 enforcement measures and the September 2021 ban.
- Thereafter it lies outside the central 95 percent range of the 40 largest comparison countries.

Largest recipients relative to GDP, 2025

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	Country	$ bn	% GDP
1	Venezuela	88.7	89.0
2	Ukraine	124.5	58.5
3	Andorra†	1.9	42.8
4	Bolivia	24.3	37.8
5	Tunisia	19.4	33.7
6	Moldova	6.3	30.8
7	North Macedonia	5.8	30.4
8	Kenya	40.1	29.4
9	Nigeria	84.7	29.2
10	Morocco	45.9	25.1

	Country	$ bn	% GDP
11	Montenegro	2.2	23.8
12	Algeria	66.3	23.2
13	Georgia	8.7	22.8
14	Jordan	12.9	20.9
15	Ecuador	27.2	20.9
16	Trinidad and Tobago	5.3	20.5
17	Belarus	18.4	19.8
18	Argentina	134.2	19.7
19	Colombia	89.2	19.5
20	El Salvador	7.1	19.0

Gross stablecoin inflows received in 2025, in percent of 2025 GDP (IMF WEO). †The only Advanced Economy in the top 20.

Largest estimated holdings relative to GDP, end of 2025

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	Country	$ bn	% GDP
1	Venezuela	5.59	5.61
2	Ukraine	8.63	4.05
3	Kyrgyz Republic	0.69	3.18
4	St. Lucia	0.05	1.85
5	Moldova	0.36	1.78
6	Tunisia	0.89	1.55
7	Kenya	1.98	1.45
8	Nigeria	4.13	1.42
9	Pakistan	5.79	1.42
10	Vietnam	6.48	1.31

	Country	$ bn	% GDP
11	Seychelles	0.03	1.29
12	Jordan	0.79	1.28
13	Algeria	3.53	1.24
14	Bolivia	0.77	1.20
15	Montenegro	0.11	1.19
16	Georgia	0.44	1.16
17	Morocco	1.97	1.08
18	Yemen	0.08	1.07
19	Belarus	0.81	0.87
20	El Salvador	0.32	0.86

Holdings are cumulative net flows since the start of the sample, in percent of 2025 GDP (IMF WEO).

In dollars, by region

Largest estimated holdings by region, end of 2025

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Region	Country	$ bn	% GDP
Africa and Middle East	Nigeria	4.1	1.42
	Algeria	3.5	1.24
	Saudi Arabia	3.4	0.27
	Region	**24.4**	**0.36**
Asia and Pacific	China	18.0	0.09
	India	7.2	0.18
	Vietnam	6.5	1.31
	Region	**58.6**	**0.15**
Europe	Türkiye	13.3	0.83
	Russia	11.0	0.42
	Ukraine	8.6	4.05
	Region	**62.6**	**0.20**
Latin America and Caribbean	Brazil	13.3	0.58
	Venezuela	5.6	5.61
	Argentina	5.1	0.75
	Region	**32.5**	**0.46**
North America	United States	13.7	0.04
	Canada	7.7	0.33
	Region	**21.5**	**0.06**

Holdings are cumulative net flows since the start of the sample. Estimates are attributions of on-chain activity and imply nothing about the legal status of stablecoin use, which several listed jurisdictions restrict or prohibit. China combines mainland China, Hong Kong SAR, and Macao SAR.

Known drivers reproduce in our data

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Stablecoins are an outside option to the domestic financial system, used when the default option is **costly**, **restricted**, or **unstable**.

- **Unstable.** One standard deviation more exchange rate volatility in an EMDE is associated with 9.7 percent larger inflows. No association in AEs.
- **Costly.** A 10 percent higher cost of traditional remittances in a corridor is associated with 1.4 percent larger stablecoin flows.

Conditional associations. They corroborate results from the Chainalysis-based literature (Auer et al. 2025) in independently constructed data.

New in this paper: the **restricted** margin (CFMs), and the dynamics around currency crises.

CFM event study: estimates

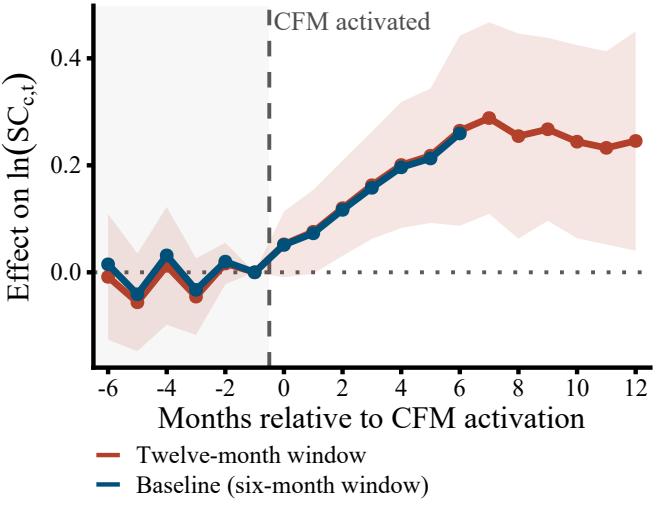
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	All CFMs	Outflow	Inflow	Neutral	Multiple
$k = -6$	0.030 (0.047)	0.015 (0.058)	0.122 (0.163)	0.016 (0.061)	−0.062 (0.072)
$k = -4$	0.032 (0.032)	0.032 (0.056)	0.004 (0.043)	0.085 (0.060)	−0.087 (0.106)
$k = -2$	0.015 (0.017)	0.020 (0.021)	0.020 (0.032)	0.043 (0.037)	−0.082* (0.050)
$k = 0$	0.038** (0.019)	0.051* (0.030)	0.066* (0.036)	0.041 (0.031)	−0.054 (0.063)
$k = 2$	0.088*** (0.029)	0.117** (0.046)	0.082* (0.044)	0.063 (0.053)	0.070 (0.128)
$k = 4$	0.118*** (0.040)	0.196*** (0.060)	0.117* (0.070)	0.051 (0.064)	0.023 (0.108)
$k = 6$	0.186*** (0.051)	0.260*** (0.089)	0.150 (0.096)	0.152* (0.080)	0.030 (0.086)
Events	217	87	43	60	27
Treated countries		48	26	41	16

Dependent variable: log stablecoin inflows. Coefficients relative to $k = -1$, selected horizons. Column 1 pools all activations. Columns 2 to 5 come from one joint regression with direction-specific paths. Event-by-event-time and event-by-country fixed effects; 283,294 observations. Standard errors clustered by country in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Outflow tightenings over a twelve-month window

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- The same directional design with the post-activation window extended to twelve months.
- Inflows stay 26 to 33 percent above controls in months seven to twelve.
- The 95 percent confidence intervals exclude zero at each of these horizons.

Outflow tightenings by type of restriction

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	Events	Average effect	Randomization p
Capital account	12	**0.241**	**0.057**
Export proceeds	8	0.242	0.171
Resident accounts	5	0.193	0.484
FX markets	6	0.134	0.574
Financial sector	17	0.110	0.379
Import payments	7	0.032	0.840
Payments and receipts	4	−0.193	0.409
Invisibles	3	1.079	0.019
Several at once	25	0.111	0.221

The 87 outflow tightenings, each attributed to the subcategory whose count rose, or to several at once (Bergant et al. 2026). Average effect over the six months after activation, from one joint regression. No subcategory is identified from more than 15 countries, so inference uses randomization tests.

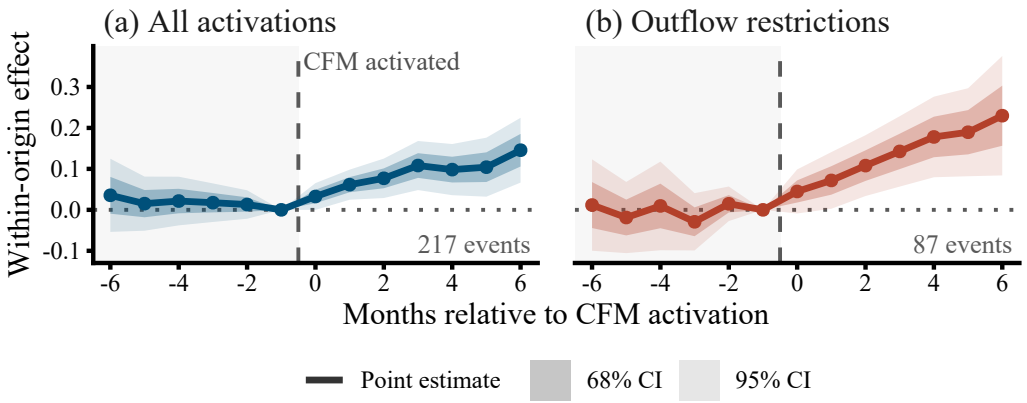
Is it the restriction, or macroeconomic stress? Back

Concern	**Robustness** (outflow effect after six months: 0.26)	
Stress before activation	No drift in inflows before activation	Table
Inflation, FX volatility	As controls: 0.19 with and without them	Figure
Crisis, sanctions	Without crisis countries: larger. Without sanctioned: similar.	Figure
Influence of outliers	Dropping each treated country in turn: unchanged	Figure

Other checks, all close to the baseline: same origin, other destinations Figure, other control groups Figure, local projections Figure

Corridor-level event study, within origin

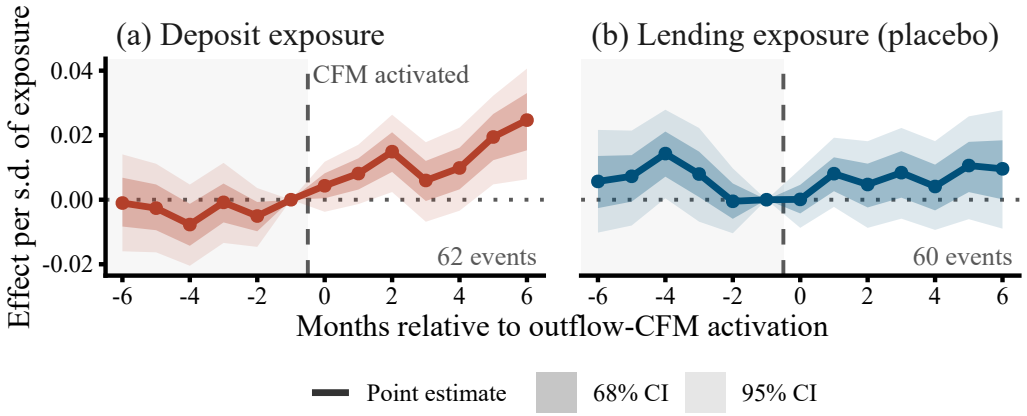
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Flows from each origin into the activating country, against the same origin's flows in the same month to countries without an activation. Global and origin-specific shocks difference out.

Bank-deposit exposure and the corridor composition of the response

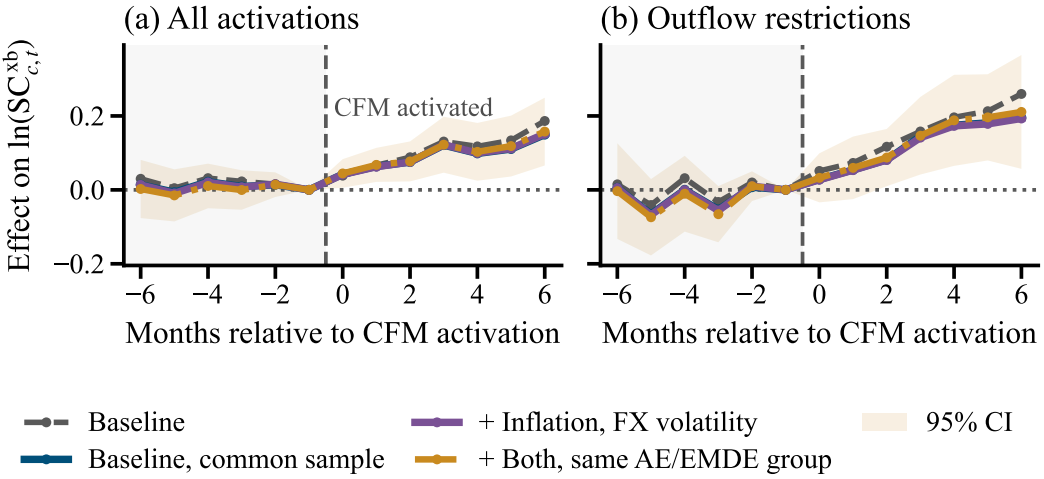
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Outflow tightenings. Exposure: the share of residents' deposits abroad held in the partner country (BIS). Country-by-month fixed effects absorb country-wide shocks. Panel (b): bank lending into the country.

Controlling for inflation and exchange rate volatility

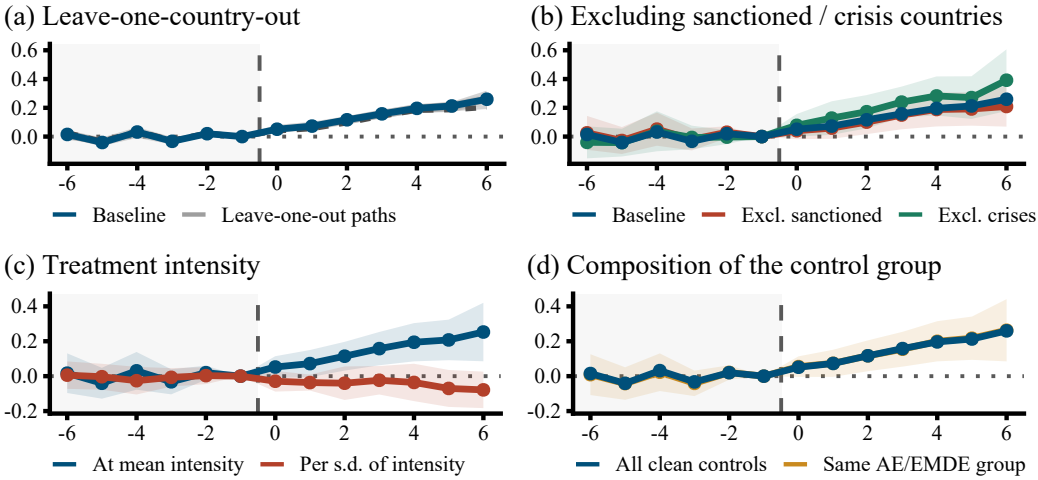
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157 countries with CPI inflation and FX volatility in every month (184 of 217 events). Six-month outflow effect: 0.195 without the controls, 0.193 with them.

Robustness of the outflow response

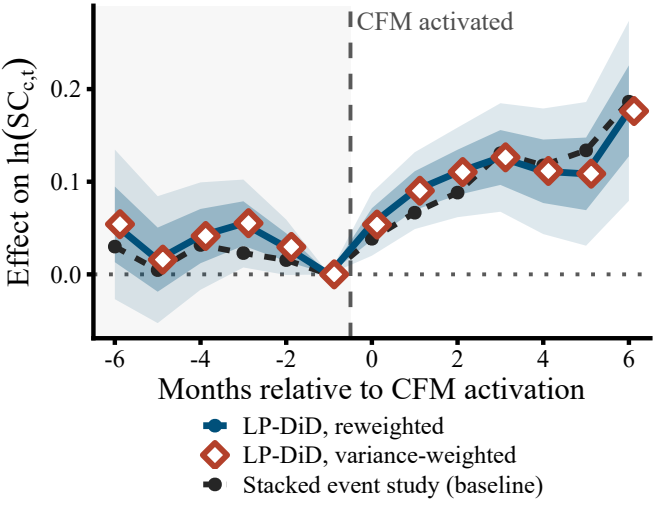
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Outflow-direction path. Crisis countries: the 28 meeting the Laeven–Valencia criterion.

Local-projections difference-in-differences

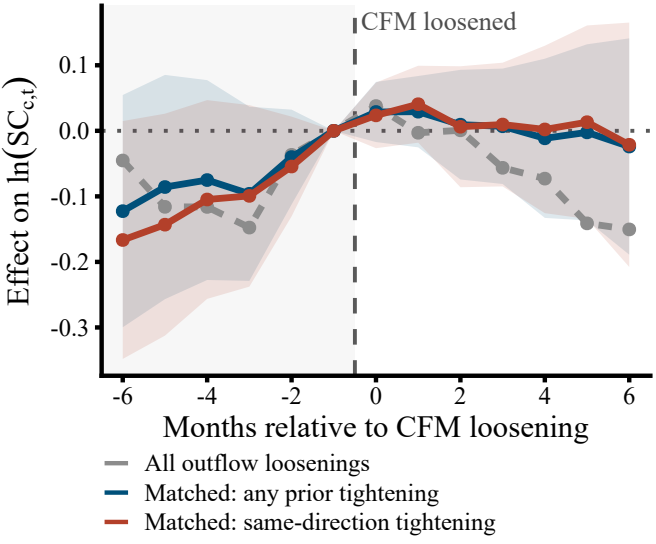
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- Dube et al. (2025). Previously treated countries never enter the controls, and the weighting of events is explicit.
- About 5 percent in the activation month and 18 percent after six months, against 3.8 and 18.6 in the stacked design.
- Variance-weighted and equally weighted estimates are nearly identical, so no subset of events drives the path.

CFM removals do not undo the response to tightenings

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- Removals after a tightening by the same country: activity stays flat.
- The late decline comes from removals no tightening precedes.
- We reject that removals fully undo the response ($p = 0.02$).

Estimates Specification

Same direction Other designs

The ratchet: estimates

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Outflow direction	Estimate	p
Effect after tightening	0.152	0.002
Effect after matched removal	0.011	0.832
Full-reversal test	0.163	0.022
Share of effect reversed	−0.07	
95% confidence interval	[−0.71, 0.57]	

Average effects from the event month to six months after. Matched removals: outflow loosenings that follow an outflow tightening by the same country. The full-reversal test asks whether the removal effect offsets the tightening effect.

Specification

The ratchet: specification

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One joint regression stacking tightenings (T) and matched removals (L):

$$\ln(SC_{c,t}) = \sum_{s\in\{T,L\}} \sum_{d\in\mathcal{D}} \sum_{\substack{k=-6 \\ k\neq -1}}^{6} \beta_k^{s,d} 1\{t - t_e = k\} \times Treat_{c,e} \times 1\{e \in s,\ d(e) = d\} + \alpha_{e,k} + \delta_{e,c} + \varepsilon_{c,t,e}$$

$d(e)$: direction of the restriction (outflow, inflow, neutral, several). L: loosenings that follow a same-direction tightening by the same country. Each event keeps its own fixed effects and clean controls; clustering by country carries the covariance between the T and L estimates.

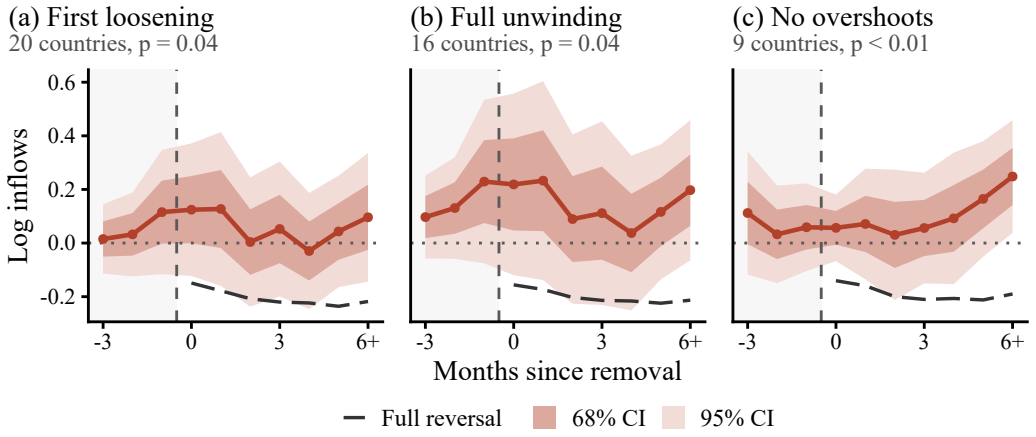
Average effect after the event, outflow direction:

$$\bar{\beta}^s = \frac{1}{7}\sum_{k=0}^{6} \beta_k^{s,\text{out}}, \qquad \bar{\beta}^T = 0.152, \quad \bar{\beta}^L = 0.011$$

Full reversal, H_0: $\bar{\beta}^T + \bar{\beta}^L = 0$. Estimate 0.163, $p = 0.022$. Share reversed $-\bar{\beta}^L/\bar{\beta}^T = -0.07$.

Full reversal is rejected in every removal design

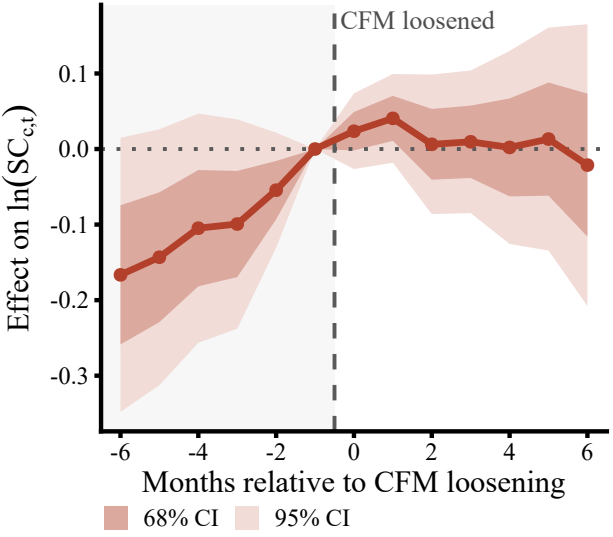
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Relative to a tightening that is kept, with the removal dated at the first loosening (a) or when restrictions are back to their pre-tightening level (b), without loosenings that go past it (c). Small-sample (CR2) p: 0.06, 0.06, 0.003.

Same-direction removals, with 68 and 95 percent bands

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- The 75 outflow removals that follow an outflow tightening by the same country. Average effect after removal 0.011 (s.e. 0.050).
- Of all 351 loosenings, 187 follow a tightening by the same country, 110 in the same direction.
- Pooled over all 351 loosenings, the effect is essentially nil. Outflow loosenings have significant pre-event dynamics, so we do not read these paths causally.
- 2021–2024 is a liberalizing period on net: 351 loosenings against 217 tightenings.

Currency crisis events

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Country	LV	FR	GDP
Chile	—	2021-07	254.1
Pakistan	2021-08	2021-08	300.4
Zimbabwe	2021-11	2021-10	40.0
Hungary	2021-11	2021-11	158.5
Lao P.D.R.	2021-11	2021-11	18.5
Sierra Leone	2021-12	2021-11	6.7
Colombia	—	2022-01	318.5
Russia	2022-03	2022-01	1,828.9
Ukraine	2022-02	2022-01	199.8
Argentina	2022-03	2022-02	486.0
Ghana	2022-02	2022-02	79.5
Egypt	2022-04	2022-03	423.3
Sri Lanka	2022-03	2022-03	88.6
Malawi	2023-09	2022-06	12.5
Botswana	2022-07	2022-07	18.8
Suriname†	2022-07	2022-07	3.1
Angola	2022-11	2022-11	83.0
Zambia	2022-12	2022-12	22.1
Belarus	2023-02	2023-01	73.7
Lebanon	2023-02	2023-02	25.7
Kenya	—	2023-04	114.7
Burundi	2023-05	2023-05	3.9
Nigeria	2023-06	2023-06	645.7
Türkiye	2023-06	2023-06	924.8
Ethiopia	2024-08	2024-08	159.8

Onset dates under the Laeven–Valencia (LV) and Frankel–Rose (FR) criteria. LV: an accelerating depreciation of at least 30 percent within twelve months. GDP is pre-crisis nominal GDP in USD billion (IMF WEO), the denominator of the outcome.

†Excluded from the baseline, which comprises 21 (LV) and 24 (FR) events.

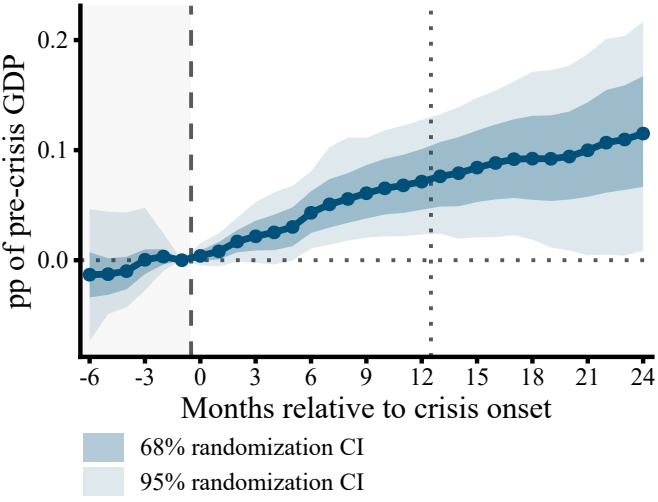
Currency crises: threats and robustness

Back

Concern	**Evidence** (baseline average effect: 0.040)
Influential events	Without Russia and Ukraine: 0.031. Equal weights: 0.041.
Dating error	Detection windows of two to six months: 0.034 to 0.042
Estimator	Callaway–Sant'Anna and Sun–Abraham agree within one standard error. Figure
CFM confound	Controlling for recent CFM activations: 0.038
Concentration	14 of 21 onsets fall between mid-2021 and end-2022. Many countries, few episodes of global stress.

Holdings around currency crises: twenty-four months

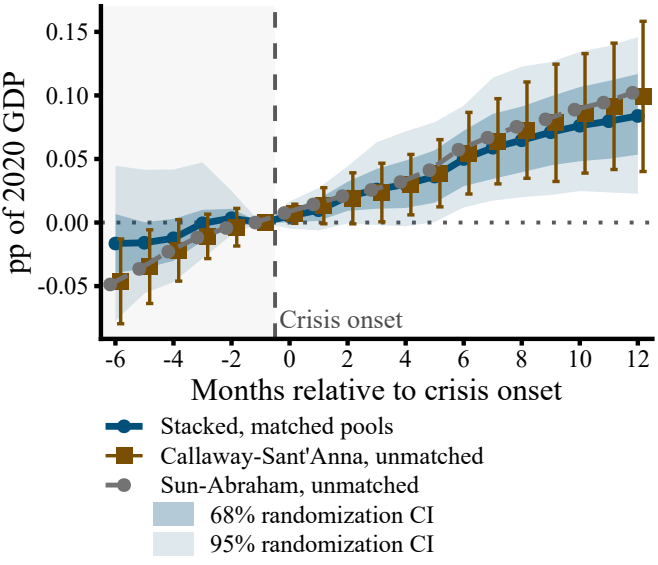
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- The identical design with the post-window extended to 24 months.
- 0.115 percentage points of pre-crisis GDP at month 24 (95% randomization interval 0.009 to 0.217).
- Accumulation continues at roughly half the first-year pace, statistically indistinguishable from a flat stock.
- Longer horizons increasingly reflect recurring crises, since each country enters once.

Alternative estimators on the unmatched panel

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- Callaway and Sant'Anna (2021) and Sun and Abraham (2021), estimated without the matched control pools.
- All three paths are on a common scale, percent of 2020 GDP.
- Both reproduce the stacked estimate within one standard error.
- On the unmatched panel, both show a drift in holdings before onset. The matched pools remove it.